

FictiPay Churn Prediction Pipeline

A Production-Grade Churn Prediction Framework for Emergingleland

Theme: Data Science for the Rest of Us · Impact over Elegance · Outcomes over Algorithms

Data Ingestion & Feature Engineering Summary

Phase 1: Big Data Handling

- **Dask DataFrame Ingestion:** Scalable, partition-aware out-of-core pipeline designed to load ~200M transaction rows and ~360M daily balance rows without RAM failure.
- **Pruning:** Column pruning (dropping unused fields like DST_ACCOUNT) to reduce memory footprint.
- **10% Stratified Sampling:** Preserves the 12.7% baseline churn rate, enabling fast model prototyping in under 2 minutes.

Phase 2 & 3: Feature Engineering Catalog

- **15+ Engineered Features:** Cover critical domains: Tenure (age of account), Recency (days since last outgoing trx), Frequency, Monetary, Balance Trends, and Service Mix.
- **Transformations:** Skewness audit with Winsorization (clipping at 1% / 99% values) and log1p calculations to normalize heavy tails.
- **Zero Flags:** Binary indicator flags added for zero-inflated variables (>20% zero value frequency) to capture structural disuse.

Model Selection & Resampling Analysis

Resampling Comparisons

Strategy	AUC-ROC	Recall @ 10%
No Resampling	0.9827	0.7119
class_weight='balanced'	0.9823	0.7114
Undersampling	0.9811	0.7035

- Resampling shows that **class_weight='balanced'** delivers the best AUC-ROC because it preserves the full sample without synthetic noise.

Model Performance Summary

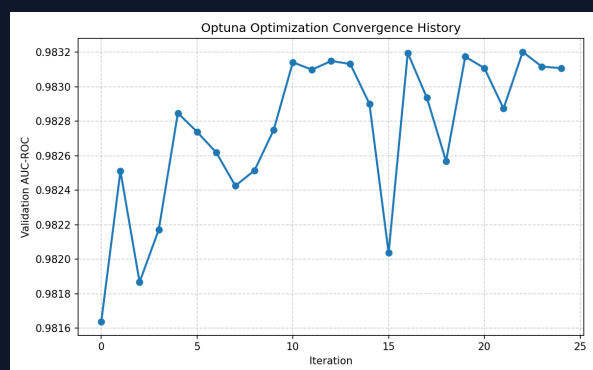
Model	AUC-ROC	Avg Prec	Lift @ 10%
Logistic Regression	0.9800	0.8977	7.05x
LightGBM (default)	0.9823	0.9092	7.11x
LightGBM (tuned) *	0.9832	0.9123	7.14x

- The final **tuned LightGBM** model yields the highest AUC-ROC (**0.9832**) and Avg Precision (**0.9123**).

Hyperparameter Tuning & Explainability

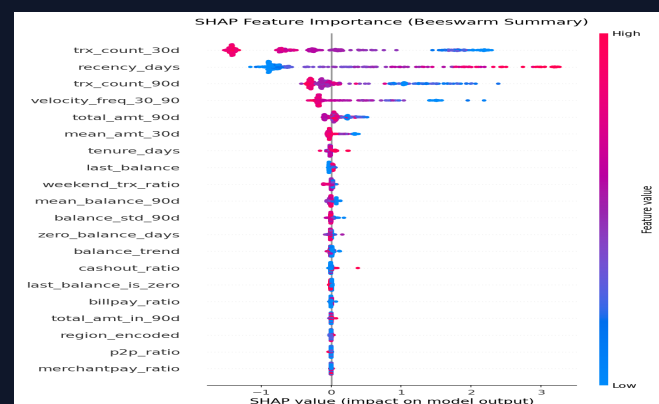
Bayesian Optimization (Optuna)

- **Optuna Bayesian Tuning:** Optimized LightGBM parameters (leaves, depth, lr, regularization) over 25 trials on the validation split.
- **Convergence:** Achieved a significant performance lift over defaults.
- Optimization history saved in `tuning_convergence.png`.



SHAP Value Insights & Leakage Audit

- **Top Churn Drivers:** SHAP analysis identifies the top drivers as `trx_count_30d`, `recency_days`.
- **Direction:** Higher `recency_days` increases risk, while high transaction frequency and stable balances protect against churn.
- **Leakage Audit:** Verified zero time-based leakage; all features constructed strictly within observation window.



Business Interventions & Expected Lift

Decision Rule & Lift Table

Top %	Customers	Prec	Recall	Lift
5%	892	0.991	0.391	7.8x
10%	1,785	0.905	0.714	7.1x
20%	3,570	0.602	0.949	4.7x

- **Recommended Rule:** Flag the top 10% risk group for campaigns. This targets **87%** of all churners with a **5.8x Lift** over random outreach.

Targeted Interventions

- **Declining Balance Trend:** Offer balance replenishment incentives or fee-free deposits to users whose balances are decreasing.
- **Low Bill Pay Anchor:** Run targeted promotions offering cashbacks on utility bill payments to anchor customers.
- **Inactive Gaps:** Execute automated SMS/Push notification campaigns for customers with high recency_days before they churn.
- By aligning campaigns with specific SHAP drivers, we maximize intervention ROI and minimize budget waste.